

**TED(10)- 4065
(REVISION-2010)**

**MODEL QUESTION PAPER (REV-10)
FIFTH SEMESTER DIPLOMA EXAMINATION IN ENGINEERING/TECHNOLOGY
(Common for BME AND MD)
THERAPEUTIC AND HOSPITAL SYSTEMS**

(Maximum Marks: 100)

TIME: 3HRS

PART - A

Answer all Questions in one or two sentences. Each question carries **2 marks**.

I.

1. List the different types of Implantable Pacemakers
2. Define a stent.
3. List the application of Incinerators
4. How pain and relief is achieved through electrical stimulation?
5. What are measured- flow Vaporizers? **5x2=10 marks**

PART - B

Answer any **FIVE** of the following questions. Each question carries **6 marks**.

II.

1. Give the principle and design of electro surgical analyzers
2. Analyze the effect of electric fields on cardiac muscles?
3. Explain the principle of intra aortic balloon pump?
4. Draw the schematic of construction of ruby laser and explain its working?
5. Draw the circuit of ground fault interrupter and give its principle?
6. Draw the schematic of Vapor compression refrigeration system and explain its working
7. Discuss the working of Centralized patient monitors?

5x6=30 marks

PART - C

Answer one full question from each Unit. Each question carries 15 marks.

UNIT I

- III** a) Describe the block diagram of defibrillator with synchronizer? **(7)**
b) Explain the principle and application of microwave diathermy? **(8)**

OR

- IV.** a) Explain the different modes of external pacemakers? **(8)**
b) Explain the principle and application of TENS? **(7)**

UNIT II

- V.** a) Explain the biological effects of radiation? **(8)**
b) What are isodose charts? Explain? **(7)**

OR

- VI** a). Draw the schematic of the distal end of the endoscope and explain? **(8)**

b) Explain LINACC? (7)

UNIT III

VII a).Explain the design of substations in hospitals? (8)

b) Explain the working of infant incubators (7)

OR

VIII a) Draw the schematic of Centralized Vacuum system in a hospital and explain (10)

b) Write short note on centralized air-conditioning system (5)

UNIT IV

IX . a) List the various flow meters and explain? (10)

b) Write short note on Video conferencing? (5)

15

OR

X a) write short notes on:

i)Bedside monitor

ii)Tele-conferencing

iii) Vaporizers

4x3=12 marks

b) List the physiological parameters adaptable to biotelemetry? (3)

15x4=60